Geophys. J. Int. (1997) **130**, 606–622

Palaeomagnetic study of the Eastern Kamchatka Cretaceous–Palaeocene island arc: new evidence concerning palaeosubduction zone absolute motion

D. M. Pechersky, M. N. Shapiro and Z. V. Sharonova

United Institute of Physics of the Earth, Russian Academy of Sciences, B. Gruzinskaia 10, Moscow 123810, Russia

Accepted 1997 February 21. Received 1996 November 15; in original form 1996 April 24

SUMMARY

An important geodynamical problem is testing for the possibility of absolute displacement of subduction zones. To consider the above problem, the most favourable geological situations are those where (1) arc-related continuous volcanism existed long enough to store significant palaeolatitude changes occurring during the accumulation of volcanic sequences and (2) a significant meridional component of the studied block displacement occurred during volcanic activity. The Upper Cretaceous and Lower Palaeocene volcanic island-arc complexes of Eastern Kamchatka, which formed on the Pacific ocean during rapid northward drift, satisfy the above conditions. Palaeomagnetic and petromagnetic investigations of the Upper Cretaceous–Lower Palaeocene volcanics of the Kumroch Ridge at 12 localities (183 hand samples) were carried out to determine the movements of the subduction zone during volcanic activity. The major magnetic mineral in all rocks is low-Ti titanomagnetite, which is typical for island-arc basalt. Most of the titanomagnetite grains from basalts are homogenous, and their Curie points are 450 °–500 °C; a small fraction of the magnetic grains have T_c = 560 °–580 °C. Most of the titanomagnetite grains in tuffs have been oxidized to magnetite. NRM stepwise thermal demagnetization was carried out, together with principal component analysis. As a result of fold, baked contact, conglomerate and 'long particles' tests, the high temperature, close to the primary, ChRM component was found. Mean palaeomagnetic directions for four stratigraphic intervals from the base to the top of the sequence are: (1) $D = 32°$, $I = 49°$, $\alpha_{95} = 3.8°$; (2) $D = 218°$, $I = -56°$, $\alpha_{95} = 5.6°$; (3) $D = 251°$, $I = -66°$, $\alpha_{95} = 2.7°$; (4) $D = 230°$, $I = -69°$, $\alpha_{95} = 7.8°$. The mean palaeolatitude changed from 30 ° to 53 ° during formation of the volcanic sequence. The drifting rate was 15–20 cm yr^{-1}, which is close to the independent estimation of the Kula plate rate for the same time. According to palaeodeclinations, the area investigated was rotated clockwise 30 °–70 ° and rocks were partly remagnetized.

Key words: palaeolatitude, palaeomagnetism, subduction.

1 INTRODUCTION

An interesting geodynamical problem is the possibility of substantial and fast absolute displacements of subduction zones. On one hand, parts of the lithosphere plunging into the mantle are often considered as anchors, and horizontal displacement relative to the surrounding mantle is physically unlikely (Cox & Hart 1986). On the other hand, as all plates undergo absolute displacements, this should also be the case for plates thrusting over troughs in subduction zones. Hence, troughs together with the dipped parts of subducting plates should also undergo similar absolute displacements. Garfunkel, Anderson & Shubert (1986) studied this problem using present-day subduction zones, and showed that the rate of their absolute displacements can be as large as 2.5 cm yr^{-1}. It is still unclear whether this value gives the maximal possible absolute rate for subduction zones or whether it characterizes only the present-day low velocities of overthrusting plates (Eurasia, America) and could have been larger in other geological epochs.

The distance between the trough and the volcanic belt changes moderately over subduction zones and usually does not exceed a few hundred kilometres. Therefore, the displacement of an ancient volcanic belt that formed over a subduction zone must reflect a migration of the subduction zone with a high degree of accuracy. If such a migration was relatively fast

© 1997 RAS

Downloaded from https://academic.oup.com/gji/article/130/3/606/674309 by Curtin University Library user on 29 April 2026

and long, and had a substantial meridional component, then variations of palaeolatitudes typical for the formation of subduction-related volcanic-sedimentary layers can be estimated by palaeomagnetic methods. Thus, for an appropriate choice of object, the minimal values of absolute displacements of subduction zones can be calculated. This paper is devoted to the study of variations of palaeoinclinations in a section of island-arc volcanic-sedimentary Late Cretaceous and Early Palaeocene rocks in the Eastern Ridge of Kamchatka (northern part of the Kumroch Ridge) (Fig. 1).

2 GENERAL GEOLOGICAL AND MAGNETOTECTONIC SETTING

Thick layers of volcanic-sedimentary rocks are widely developed in the axial part and on the western slopes of the Eastern Ridge of Kamchatka; they are dated as Upper Campanian, Maastrichtian, and probably Lower Palaeocene. In the southern part of the ridge they are distinguished as the Valaginsky series, and in the northern part as the Khapitsky Formation (Review of the tectonic evolution of Kamchatka 1987) (Fig. 2). Layered reworked basalt, andesite and more rare picrite tuffs dominate in their composition. Usually, an alternation of rocks containing fragments of different sizes, from agglomerates and breccia to sandy and muddy tuffs, is observed. Tuffites, in which pyroclasts are mixed with clayey and flinty material, are widely developed. Volcanic members and lenses are common in the sections where coarse-grained tuffs dominate. Konstantinovskaya *et al.* (1993) consider this complex as a volcanic flinty-tufaceous formation that was accumulated on the slopes of an ensimatic island arc. Low-K island-arc tholeiites prevail in the basalts of the complex, while the fraction of calc-alkaline rocks is relatively small (Magakian *et al.* 1993); picrites are assumed to belong to the boninite series (Kamenetsky *et al.* 1993). Taken together, this allows us to compare these units with the volcanics of primitive oceanic island arcs (Zinkevich & Tsukanov 1993).

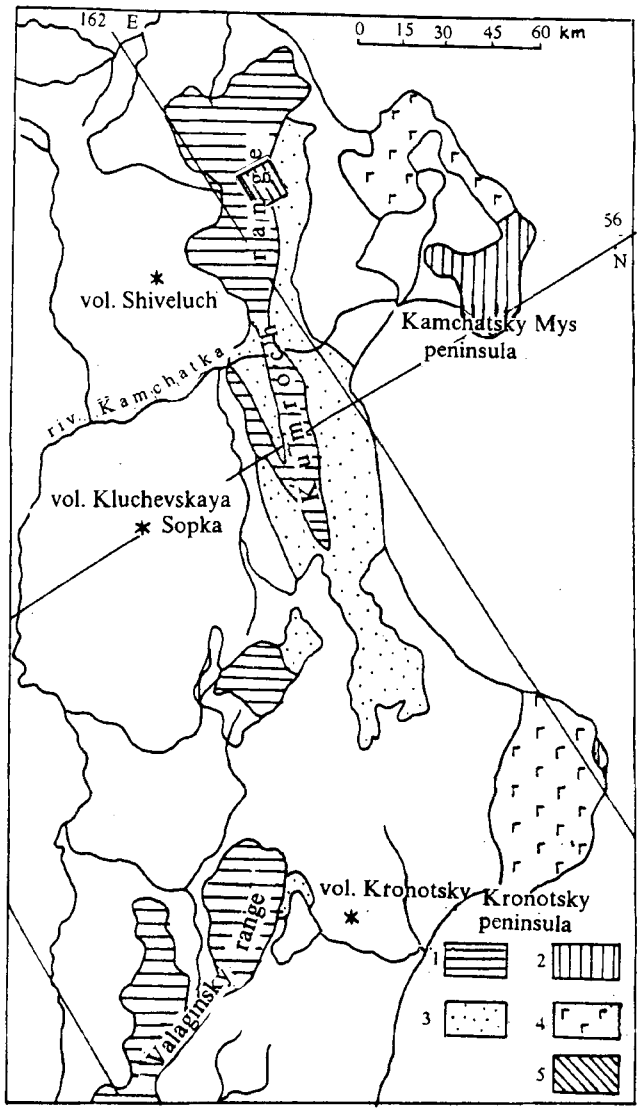


Figure 1. Schematic map of the East Kamchatka Upper Cretaceous (1, 2) and Lower Palaeogene (3, 4) complex distributions. 1, 3 = East Ridges; 2, 4 = East Peninsulas; 5 sampling area.

These volcanic-sedimentary layers were highly deformed, resulting in a series of steep imbricated thrusts of southeast vergence. Occasionally, small gentle overthrust sheets were formed. The complicated tectonic structure together with sharp facies variations and rare palaeontological occurrences prevents us from making a detailed stratigraphic subdivision of the complex. The bottom of the complex does not crop out anywhere, and its upper part transforms concordantly to terrigenous flysch layers of the Talnikovsky (Valaginsky Ridge) and Drozdovsky (Tumrok and Kumroch ridges) Formations. A large fraction of radiolarians from the flinty rocks of the Valaginsky and Khapitsky Formations were determined to be of Upper Campanian to Maastrichtian age (Tsukanov 1991). The Upper Palaeocene plankton foraminifera were found in the lower part of the Talnikovsky flysch in the Valaginsky Ridge (Bakhteev *et al.* 1994). Thus, the upper part of the Valaginsky series could be of Lower Palaeocene age.

Similar structures in Upper Campanian, Maastrichtian and Lower Palaeocene volcanic sedimentary complexes also developed on the eastern slopes of the Sredinny Ridge (the Kirganiksky Formation) (Flyorov & Koloskov 1976) and in the southern part of the Koryak highland (the Achaivayam Formation) (Alekseev 1979). Many researchers consider these complexes, together with the complexes of the Eastern Ridge, as a single palaeoisland-arc system, which we name the Achaivayam–Valaginsky palaeoarc (Shapiro 1995). It is assumed that it was formed as an intra-oceanic arc, and later, due to northwestward drift, became part of Eurasia. It is likely that the Vatyn–Lesnovsky overthrust is the suture that was formed as a result of this collision; along this suture the Upper Cretaceous volcanic sedimentary complexes overthrust coeval terrigeneous series of the western Kamchatka and Ukelayatsky trough (Alekseev 1979; Shantser *et al.* 1985). The overthrust is overlapped by shelf sediments of the second half of the Middle Eocene and intruded by slightly older granites (Shantser *et al.* 1985).

The palaeomagnetism of the Upper Cretaceous–Palaeocene formations of the eastern ridges of Kamchatka has only been considered in one paper (Gurevich & Surkis 1994). According to that data, the ChRM of the studied rocks was mainly formed simultaneously with folding, and the palaeolatitude of the sites studied varies from 20° to 48°N. According to palaeomagnetic data, the palaeolatitude of the Upper Cretaceous island-arc complexes of the Olutorsky zone varies

Downloaded from https://academic.oup.com/gji/article/130/3/606/674309 by Curtin University Library user on 29 April 2026

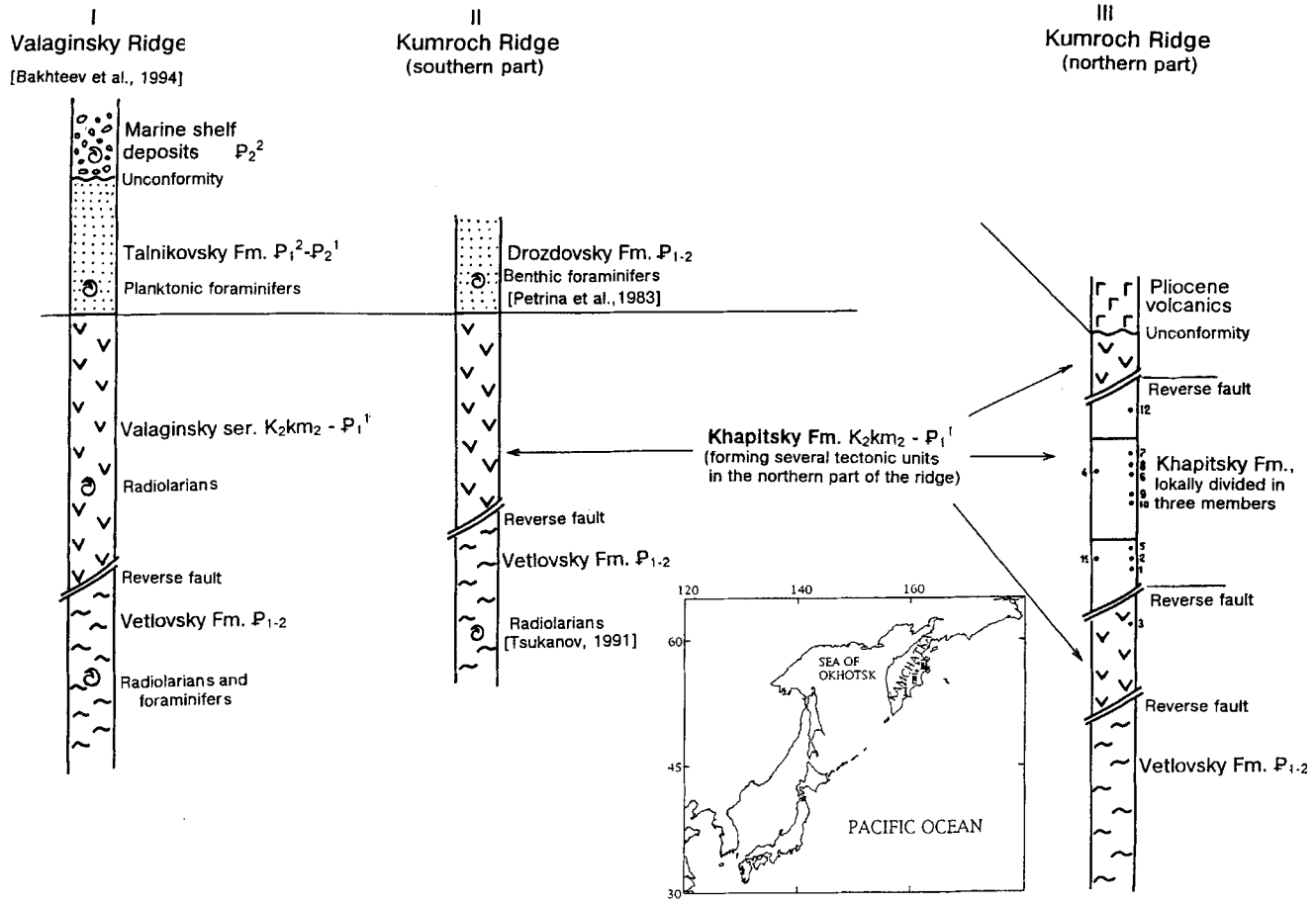


Figure 2. Stratigrapic scheme of Kamchatka's Upper Cretaceous and Lower Palaeogene Eastern Range. Short lithological descriptions of the stratigraphic units can be found in the text. The positions of the palaeomagnetic sites (1–12) are shown in the right-hand column. The locations of the sections (I, II, III) are shown on the map.

from 36° to 47°N (Kovalenko 1990, 1992; Savostin & Heifets 1988). Rapid displacement occurred in the Palaeocene (Kovalenko 1992).

Thus, the available palaeomagnetic data prove the hypothesis that the Achaivayam–Valaginsky palaeo-arc drifted substantially northwards. This drift had to finish prior to the Middle Eocene, the upper age limit of the Vatyn–Lesnovsky thrust. If the arc drifted after the formation of the volcanic belt, then the drift was not associated with a displacement of the subduction zone. If, even partly, the drift occurred simultaneously with volcanism, then the subduction zone would have had to move relative to the underlying mantle. The first hypothesis seems to be unlikely, because no essential manifestations of the Late Palaeocene and Early Eocene oversubduction volcanism are known in Kamchatka or on the Koryak highland to the northwest of the Achaivayam–Valaginsky arc; hence, no subduction zone existed in this area, where the oceanic crust, which separated the arc and the continental margin in the early Palaeocene, was consumed.

When considering the second hypothesis, one has to pay attention to the fact that, in the second half of the Campanian, in the Maastrichtian and Lower Palaeocene, the Achaivayam–Valaginsky arc was closest to the northeastern Asia zone where thick subduction-related series were formed. If the subduction zone was inclined at that time towards the continent, the arc could not have moved relative to it. Such a displacement would have been possible only if the subduction zone had the opposite inclination, and the oceanic lithosphere that was part of the Eurasian (or the North American) plate was consumed there. Immediately afterwards, the dipped part of the plate had to move towards the continent with an absolute rate equal to the rate of the most northwestern oceanic plate of the Pacific ocean. The Achaivayam–Valaginsky arc, which was formed on an oceanic plate, had to move with the same rate. In a large number of palaeoreconstructions proposed for the Late Cretaceous and Early Palaeocene, the Kula plate is considered the most northwestern; at that time its absolute rate was as large as 20 cm yr^{-1}, with a northern component of about 10–15 cm yr^{-1} (Engebretson, Cox & Gordon 1985; Kononov 1989). If the arc moved northwards with the same rate, then in less than 10 Myr a difference in palaeolatitudes of accumulation could become large enough to be detected by palaeomagnetic methods. The purpose of this paper is to test this assumption.

3 GEOLOGICAL CHARACTERISTICS OF THE SITES OF THE STUDY

The Upper Cretaceous–Lower Palaeocene rocks form the axial part and the western slopes of the Kumroch Ridge, where they are considered as the Khapitsky Formation. In the southern part of the ridge, the formation is concordantly overlapped by

Downloaded from https://academic.oup.com/gji/article/130/3/606/674309 by Curtin University Library user on 29 April 2026

terrigenous flysch of the Drozdovsky Formation (which does not contain tufaceous material). Small complexes of radiolarian and benthic foraminifera as well as rare fragments of Inoceramus shells were found in different parts of the Khapitsky Formation, dating it as Upper Campanian–Lower Palaeocene (Tsukanov 1991). At the Valaginsky Ridge, formations analogous to the Khapitsky Formation (the Valaginsky series) are overlapped by the terrigenous flysch of the Talnikovsky Formation (analogous to the Drozdovsky Formation) which contains plankton foraminifera of the Upper Palaeocene (Bakhteev *et al.* 1994).

The Cretaceous–Palaeocene formations of the Kumroch Ridge form series of sheets with complex internal structure and are strongly deformed. Steeply inclined or overturned folds are mapped in the southern part of the ridge, and monoclines in its northern part. From the east, the entire system is bordered by a thrust fault, along which the Khapitsky Formation overthrust the cherty-clayish Palaeocene–Lower Eocene Vetlovsky Formation. The age of the deformation is unknown; however, to the south, in the Valaginsky Ridge, analogous thrust faults and folds are unconformably overlapped by the Middle Eocene sediments (Bakhteev *et al.* 1994).

To select orientated samples, one of the thickest thrust sheets was taken in the northern part of the Kumroch Ridge (Fig. 3). Here the Khapitsky Formation is subdivided into three conformably bedded members. The lowest member, with a thickness of up to 1200 m, is formed by green-grey tufaceous siltstones, sandstones and gravel tuffs deposited in water (tephroids). The azimuths and dips of the layers vary greatly (Table 1). The rocks have many cracks, but the cracks are not of the cleavage type, i.e. no regular reorientation of grains has occurred. Samples were selected from four sections (Table 1, Fig. 2). Sections 1, 2 and 5 refer to the upper part of the lowest member and have similar stratigraphic positions. The exact stratigraphic position of section 11 with respect to the other three sections was not determined.

The amount of coarse-grained tuffs gradually increases up through the lowest member, and it transforms gradually into the middle member with a thickness of 2500–2800 m. This member is composed of agglomerate basalt tuffs and tufaceous conglomerates, alternating with thin layers of fine-grained tuffo-terrigenous stratified rocks. Three basalt sills were found there; their contacts with the host rocks were hot. The sills must have formed at about the same time as the host rocks, because similar rocks are widely found in conglomerate pebbles both above and below these bodies. In the middle member, samples were taken from various parts of the stratified tuffo-terrigeneous rocks and from three sills: from bottom to top, these are sections 10, 9, 6, 7 and 8 (Table 1, Figs 2 and 3). For testing, pebbles similar to the rocks from the sills were sampled from one of the tuffo-conglomerate horizons.

The number of layers of fine-grained rocks increases gradually to the upper part of the middle member, and at the level where these rocks form more than one-third of the section, a conventional boundary between the middle and upper members is located. The upper member consists of an irregular

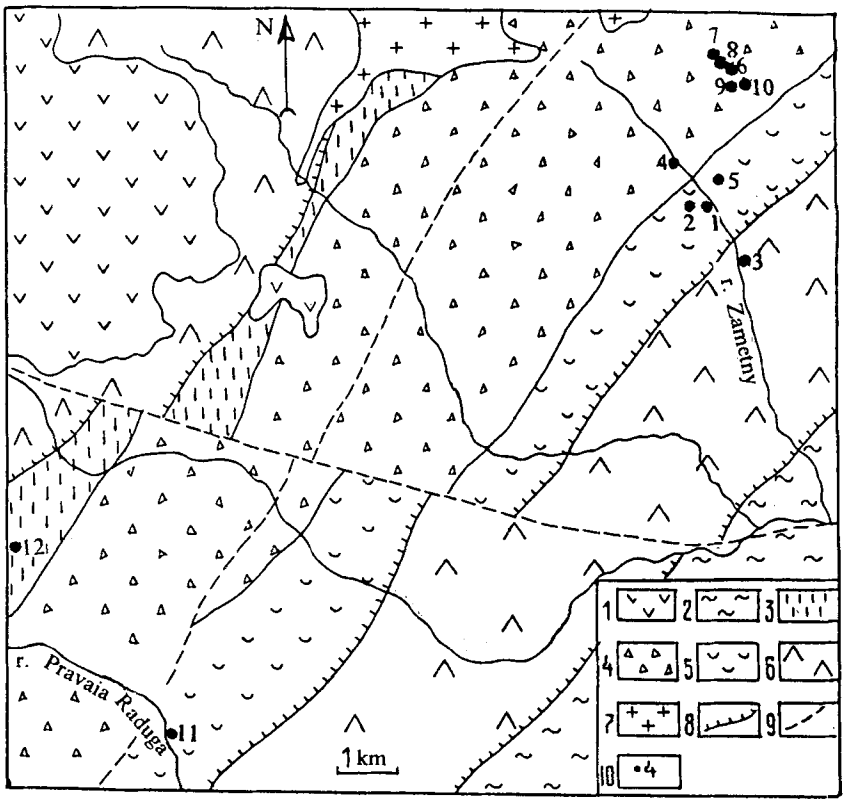


Figure 3. Geological schematic map of locations of research sections in the northern Kumroch Ridge. 1: the Pliocene volcanics; 2: the Palaeocene–Lower Eocene cherty-clayish deposits (the Vetlovsky Formation); 3–6: the Khapitsky Formation; 3: the uppermost member; 4: the middle member; 5: the lowest member; 6: non-dismembered part; 7: the Pliocene diorite-porphyrites; 8: major thrust faults; 9: other faults; 10: locations of sections and their numbers.

Downloaded from https://academic.oup.com/gji/article/130/3/606/674309 by Curtin University Library user on 29 April 2026

Table 1. Some characteristics of sampled sites of the Khapitsk Formation.

Site	Stratigraphic position	Rocks	Thick., m	N	Bedding Azimuth	Dip
12	Upper member	Tuffs	160	18	325 (312–334)	76 (58–82)
8	Middle member	Tuffs	20	3	352	78
7	Middle member	Basalts (sill)	5	4	330	75
7	Middle member	Tuffs	4	5	337	73
6	Middle member	Basalts (sill)	10	11	339	65
6	Middle member	Tuffs	4	16	339	62
9	Middle member	Tuffs	8	8	325 (322–328)	64 (48–69)
10	Middle member	Basalts (sill)	9	3	345	84
10	Middle member	Tuffs	1	2	345	84
1	Lower member	Tuffs	32	32	315 (302–331)	53 (43–72)
2	Lower member	Tuffs	23	26	312 (306–318)	72 (66–83)
5	Lower member	Tuffs	27	24	274 (271–285)	60 (54–68)
11	Lower member	Tuffs	78	14	283 (273–301)	80 (73–86)
3	Non-dismembered	Tuffs	22	17	315 (307–324)	115(103–127)

Thick. = the thickness of studied interval of section, m; N = number of hand-samples.

intercalation of massive coarse-grained tufaceous-sedimentary rocks and thin flysch-like layers of tuffo-sandstones and tuffo-siltstones. The upper part of the section is poorly exposed. Also, a noticeable part of the upper-member sections is in the vicinity of the Pliocene intrusives. This is why orientated rocks were sampled from intercalated tufaceous sandstones and siltstones of one section in the lower part of the upper member (section 12, Table 1, Figs 2 and 3).

Numerous thin flinty bands are typical for the sections of the lower member. These bands as well as fine-grained tuffites often contain recrystallized radiolarian skeletons. This indicates that the layer was formed at the bottom of a sea basin, and thus the primary dip of layers could not exceed 1°–2° there.

Section 3 is part of another thrust sheet; it is separated by a steep overthrust parallel to the bedding but can be clearly distinguished because of an intensive fracturing and mylonitization of rocks in the 20–30 m zone. The Khapitsky Formation changes facies southeast of this zone, and cannot be subdivided into the same three layers as to the northwest of the thrust zone. It differs from the rocks of the lowest member by the overturned bedding (Table 1), structural and textural characteristics and the degree of tuff zeolitization; usually it is considered as the lowermost part of the Khapitsky Formation. Cherty tuffites interlayered with tufaceous siltstone and sandy tuffs were sampled.

4 METHOD

Hand samples from the outcrops were sampled manually and were orientated by magnetic compass. Each hand sample was cut into 1–3 cubic specimens with 2 cm ribs. Specimens were subjected to stepwise thermal demagnetization in a furnace with a four-layer permalloy shield. After each heating, the natural remanent magnetization (NRM) was measured on a JR-4 spin-magnetometer within large Helmholtz coils, and the magnetic susceptibility (k) was measured on a KLY–2 kappameter. k was measured not only for petromagnetic information but to control changes in the samples during laboratory heating.

The results of the palaeomagnetic measurements were processed by NRM component analysis, great-circle analysis (Halls 1978), Graham conglomerate and fold tests, the reversal test, and the long-particle test (Pechersky 1970). The idea behind the latter is to measure the orientation of elongated grains of opaque minerals in the plane of the layer in orientated thin sections. The orientation pattern of elongated grains must correspond to the palaeomagnetic declination that existed during sedimentation. To exclude grain orientations caused by processes such as currents during grain deposition and sediment deformation, the orientations of non-magnetic elongated grains were measured.

To estimate the reliability of the palaeomagnetic measurements, the fraction of viscous remanent magnetization (ChRM/NRM) and the Königsberger ratio (Q_n) were taken into account; magnetic carriers were studied for 43 samples representing various rock types: saturating magnetization J_s, remanent saturating magnetization J_{rs}, and remanent coercivity H_{cr} were measured. Thermomagnetic analysis of $J_{rs}(T)$ and $J_s(T)$ was carried out on thermomagnetometers with automatic registration designed by Yu.V.Vinogradov.

For basalt samples from sills and samples of magnetic tuffs, the composition and structure of grains of opaque minerals were studied on a Camebax electron microprobe. A probe of 1–2 μm in diameter was usually used; if the studied grains were heterogeneous their average composition was measured with a probe of 10–20 μm. The Ti content and/or Ti/Fe ratio in titanomagnetite was used for the computation of the expected Curie temperature.

5 DATA OF THE ELECTRON MICROPROBE ANALYSIS (TABLE 2)

The average content of Ti in titanomagnetite from basalt sills is low and varies greatly from one sample to the next, which is typical for igneous rocks from island arcs (Pechersky *et al.* 1993). The composition of titanomagnetite is usually homogeneous within grains; the content of TiO_2 in the central part of some grains is 10 per cent and increases to 15 per cent at the grain edges; this phenomenon is typical for crystallization in an oxygen-closed system (Pechersky *et al.* 1993). Grains of titanomagnetite are structurally homogeneous: decomposition

Downloaded from https://academic.oup.com/gji/article/130/3/606/674309 by Curtin University Library user on 29 April 2026

Table 2. Results of the electron microprobe analysis.

Rocks	N	Size	TiO_2	FeO*	MgO	MnO	Cr_2O_3	Al_2O_3	T_c*, °C
Tuffs									
L260	15	4–50	0.0	92.6	0.2	0.0	0.0	0.1	580
				91.2–93.8	0.0–0.8			0.0–0.8	
L233	12	10–40	5.0	87.0	0.8	0.3	0.0	1.9	495
			4.1–6.7	86.0–91.5	0.1–1.6	0.1–0.4		1.2–4.0	460–508
L349	10	6–20	6.1	88.1	0.3	0.4	0.0	1.9	470
			2.5–12.6	75.2–91.1	0.1–0.5	0.1–0.9		1–4.5	345–542
L350	10	20–70	7.0	85.1	0.3	0.9	0.0	0.4	455
			5.5–14.1	70.0–87.1	0.1–0.6	0.1–2.2		0.1–2.2	320–480
Basalts									
Sh127	10	20–50	13.1	80.7	0.1	0.1	0.2	1.5	385
		9.6–15.0	75.1–84.2	0.0–0.3	0.0–0.8	0.0–0.5	0.1–1.7	300–405	
Sh151	10	10–200	5.8	86.8	0.2	0.2	0.1	1.8	475
			4.9–7.6	85.5–88.0	0.1–0.5	0.1–0.3	0–0.3	1.1–3.0	450–490
L283	10	20–400	4.4	89.3	0.3	0.4	0.5	2.2	500
			0.0–9.3	87.1–91.1	0.1–0.5	1–1.0	0.1–1.0	0.1–4.0	410–580

N is the number of measured mineral grains; Size = the size of measured grains in μm; oxide contents are in weight per cents; FeO* is the content of iron calculated as a ferrous oxide; T_c* is the Curie temperature calculated from Ti content or Ti/Fe ratio. The upper rows refer to average values, lower rows to the limits.

structures were not found at a magnification of more than 10 000×.

The composition of titanomagnetite in tuffs and sills is essentially the same. This indicates that they originated from the same eruptive source. Despite compositional similarity, grains of titanomagnetite in tuffs are less homogeneous than in basalts; however, at a magnification of more than 10 000× decomposition structures were not found. Along with grains of titanomagnetite, grains of magnetite were found (sample L260); however, even in these the content of TiO_2 at the edges may be as high as 5–10 per cent.

6 PETROMAGNETIC (ROCK-MAGNETIC) PROPERTIES OF BASALTS AND TUFFS

According to thermomagnetic analysis, the major magnetic mineral in the basalt sills is low-Ti titanomagnetite. Its T_c and T_d are 490 °–510 °C (Table 3). Judging from the small range

Table 3. Some petromagnetic properties of basalts and tuffs.

Rocks	AB	N	J_s	J_{rs}/J_s	H_{cr}	T_d	T_c	J_{rst}/J_{st}	J_{st}/J_s
Basalts									
Sills	A	3	2.22	0.04	18	480	510(1)	1.6	0.93
			2.05–2.32		14–21	450–500		1.1–2.0	
Sills	B	5	2.22	0.05	14	490	500	1.9	0.88
			1.86–2.51	0.03–0.07	10–17	480–500	490–510	1.5–2.4	
Pebbles	3	2.59	0.08	11	500	–	2.6	–	
		2.07–3.04	0.04–0.11	8–14	–		1.6–3.2		
Tuffs									
Exocon	A	1	1.64	0.18	60	580	570	0.82	0.71
Heating	A	6	0.66	0.12	50	585	–	0.76	–
			0.27–0.91	0.09–0.16	40–60	580–600		0.6–1.0	
HM	A	9	1.02	0.05	32	565	575	1.25	0.87
			1.16–2.42	0.03–0.10	21–52	540–580	560–590	0.8–1.8	0.8–0.95
HM	A+B	3	2.82	0.05	35	560?	–	1.94	–
			2.64–3.0	0.02–0.07	24–52			1.6–2.5	
HM	B	4	1.55	0.07	37	570	580	2.6	0.87
			0.46–2.22	0.03–0.1	27–51	560–580		3–4.4	0.8–0.95
LM	A	3	0.13	0.007	38	545	4.2	–	
			0.10–0.15	0.006–0.008	27–53	520–560		3.6–5.1	
LM	B	6	0.13	0.005	36	580	–	8.4	–
			0.04–0.17	0.001–0.011	31–41	–		1.7–16.4	

Exocon = baked tuff; heating = zone of tuff heating near the sill; HM = high-magnetic rock, LM = low-magnetic rock; A,B refer to the samples with magnetization A or B; N is the number of samples; J_s is saturation magnetization, Am^2kg^{-1}; J_{rs} is saturation remanent magnetization; H_{cr} is remanent coercivity, mT; T_d is deblocking temperature of J_{rs}; T_c is the Curie temperature of J_s; J_{st} and J_{rst} are saturation magnetization and saturation remanence after sample heating to 500 °C (basalts) and 600 °C (tuffs). The upper rows refer to average values, lower rows to the limits.

Downloaded from https://academic.oup.com/gji/article/130/3/606/674309 by Curtin University Library user on 29 April 2026

of J_s variations, the concentration of this mineral is uniform and is about 2–3 per cent. Low values of J_{rs}/J_s (0.02–0.05) and H_{cr} (10–14 mT), and highly concave J_{rs}(T) curves (Fig. 4) indicate the presence of coarse multidomain grains. For some samples the J_{rs}(T) curves showed that, besides $T_d < 500\,°C$, magnetite ($T_c = 580\,°C$) was found. The absence of the T_c of magnetite in the J_s(T) curves and higher values of H_{cr} observed for these samples indicates a low concentration and very small size of magnetite grains.

The calculated Curie points in basalts Sh151 and L283 (475 ° and 495 °C) coincide with the measured values. Hence, homogeneous grains of titanomagnetite with a composition close to the primary one prevail in these samples. In sample Sh127, which contains heterogeneous grains, the calculated $T_c^* = 300-405\,°C$ is less than the measured $T_c = 490\,°C$.

Thermomagnetic analysis revealed that the major magnetic mineral in tuff samples is magnetite ($T_c = 570–590\,°C$). The values of k and J_s are widely scattered (Tables 3 and 4). The content of magnetite in magnetic tuffs is close to the titanomagnetite content in basalts; it indicates that the source of material was close to the place of its deposition. This conclusion is also supported by the large amount of coarse angular clastic grains of opaque minerals in magnetic tuffs. Such tuffs, enriched by magnetite, are interlayered with tuffs with a low magnetization, where the content of magnetite is less than 0.1 per cent. Rocks with low magnetization prevail in the lowest member; highly magnetized rocks occur in the middle and upper members. This pattern is likely to reflect non-uniformity with time of the volcanic activity. Compared to basalts, tuffs have higher values of H_{cr} (Table 3).

The calculated Curie points of tuffs and basalts are the same (Table 2); however, unlike basalts, the calculated and measured Curie points of tuffs disagree. The discrepancy between the calculated and measured Curie points, the high coercivity and predominance of large grains (according to microscopic study) indicate that titanomagnetite has undergone more intensive changes in tuffs than in basalts. The absence of decomposition structures at a magnification of more than 10 000 × can be explained either by a single-phase oxidation [$z = 0.6–0.8$ (Nishitani & Kono 1983)], or by very thin low-temperature decomposition (lamellae less than 0.1 μm wide) which cannot be detected by electron microanalysis. Both of the processes are possible when debris is deposited in water where it is oxidized at low temperatures. Occasionally, after heating to 400 ° – 500 °C a decrease in k, J_s and J_{rs} to 0.7–0.8 of their initial values was found; we connect this with the maghemitization. As follows from the results of the NRM thermal demagnetization ($T_d = 580\,°C$ and sometimes higher), the major carrier of the high-temperature NRM component (ChRM) in basalts and tuffs is usually magnetite; at the same time $T_c = 490\,° – 510\,°C$ in basalts. Such a discrepancy indicates the presence of a large amount of fine-grained magnetite in basalts, which makes a major contribution to the NRM while its influence on J_s and J_{rs} is negligible. This conclusion is supported by the behaviour of NRM and J_{rs} during AF-demagnetization (Fig. 5). The stable part of the NRM with

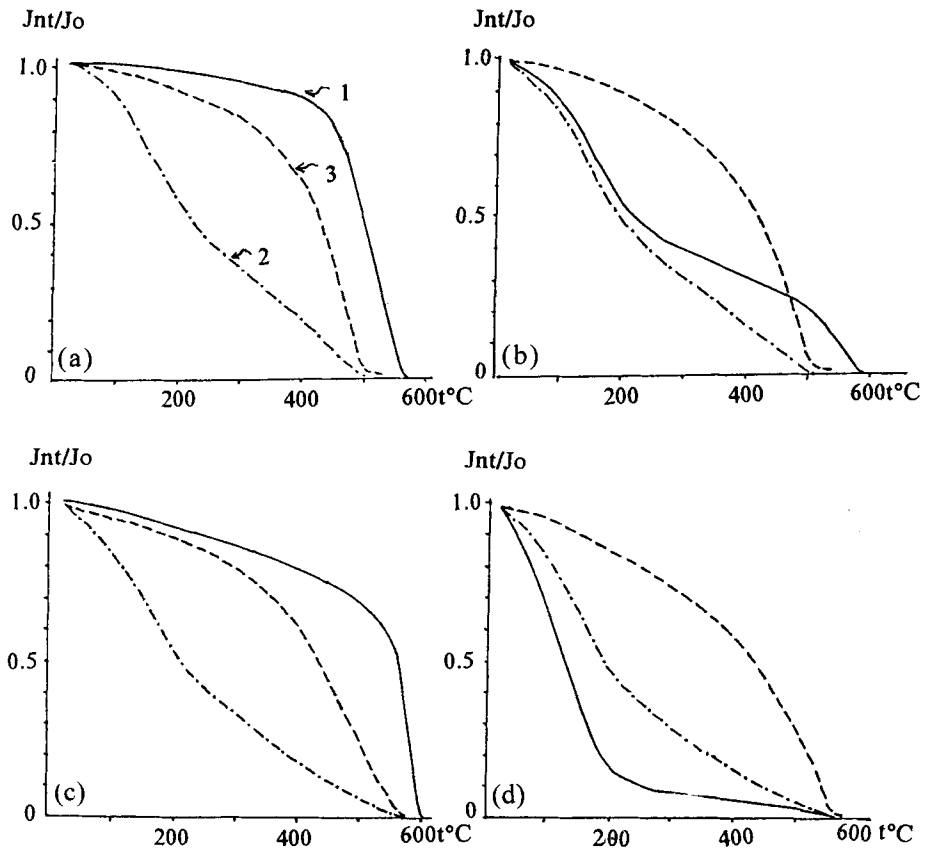


Figure 4. Typical normalized thermomagnetic curves of NRM(T) (1), J_{rs}(T) (2) and J_s(T) (3). (a) Sample Sh130, basalt; (b) sample Sh285, basalt; (c) sample L134, tuff; (d) sample L333, tuff.

Downloaded from https://academic.oup.com/gji/article/130/3/606/674309 by Curtin University Library user on 29 April 2026

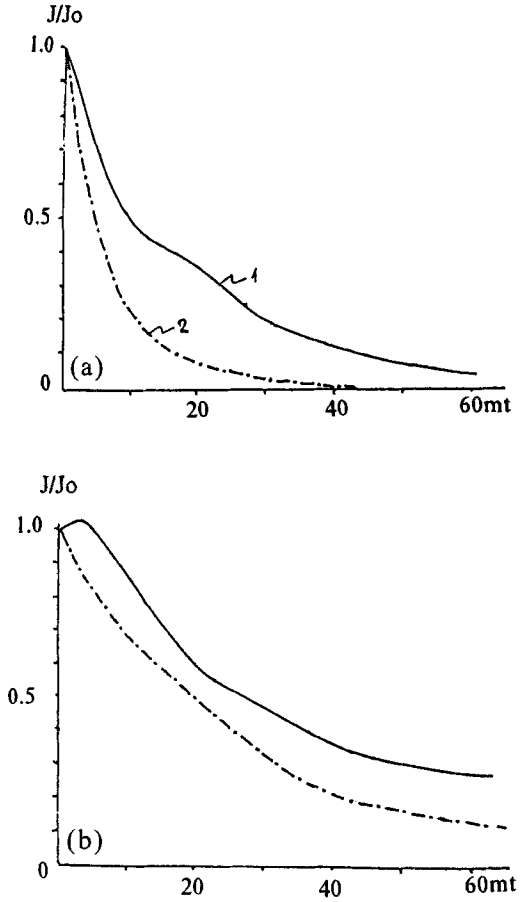


Figure 5. Typical normalized AF-demagnetization curves for NRM (solid line) and J_{rs} (dot-dashed line). (a) Sample Sh151, basalt; (b) sample L234, tuff.

$T_d > 500\,°C$ is associated with magnetite fine grains. The unstable part of the NRM, which has an essentially viscous origin, is associated with large grains of titanomagnetite, magnetite and maghemite (Fig. 4, Table 4).

It is well known that in basalts magnetite is a product of high-temperature heterophase oxidation of primary magmatic titanomagnetite. Such magnetite often has primary thermoremanent magnetization (TRM). Tuffs are the result of the redeposition of the products of basaltic volcanism; that is why they contain mainly clastic magnetite, with which primary detrital remanent magnetization (DRM) should be associated. This notion is supported by the ratio of the NRM and the anhysteretic remanence (ARM). Basalt Sh127 (A magnetization) has NRM/ARM $=1.1$, which does not contradict with the TRM nature of the NRM; basalt Sh151 (B magnetization) has NRM/ARM $=0.1$, which may be explained by a large viscous component in the NRM. Four tuffs with magnetization A have NRM/ARM in the range 0.09–0.3; such values are typical for the DRM. Two tuffs with magnetization B have NRM/ARM $=0.05$; this value may be explained by a large portion of viscous remanence.

7 RESULTS OF THE PALAEOMAGNETIC STUDY

The results of the NRM thermal demagnetization and component analysis allowed us to distinguish middle to high-temperature components (ChRM), which can be divided into two groups (Figs 6, 7 and 8): A—with a northeastern declination of the normal polarity and a southwestern declination of the reversed polarity (Figs 6a–d), and B—with a southeastern declination of the reversed polarity (Figs 6h–i). In some samples, both components are present in the NRM (Figs 6e–g). In the latter case, the deblocking temperature for the B component is lower than that for the A component. The results were grouped according to their stratigraphic position.

7.1 The lower member

7.1.1 Section 5 (Table 4)

Thermal demagnetization allowed the detection of ChRM components of reversed and normal polarity on orthogonal plots (Fig. 7, Table 5). In the ancient coordinate system the directions of unit vectors can be divided into two groups (magnetizations A and B, Fig. 7). For magnetization A, $K_a/K_c \approx 1$; for B, $K_a/K_c = 0.51$ (Table 5). This means that magnetization B can be post-folding or synfolding. In 10 samples, both magnetizations were found on orthogonal plots, and magnetization B has a lower T_d than A (Fig. 6g). Eight samples have only B magnetization (except for the low-temperature viscous magnetization).

7.1.2 Section 1

This section is located stratigraphically higher than section 5. This is proved by a decrease in the amount of magnetic minerals in tuffs (Table 4) and a reduction in the layers with a high magnetization in the section to less than one-third of its thickness. Similar in section 5, the high-temperature components form two groups (A and B, Table 5, Fig. 7) against the backgrounds of greatly scattered single ChRM directions. For both A and B magnetizations $K_a/K_c \geq 1.5$. In some samples both magnetizations were found, T_d being lower for B magnetization (Figs 6e and f); in many samples only B magnetization was present (Fig. 6 g). In this case the great-circle analysis is not effective.

7.1.3 Section 2

This section is similar to section 1, stratigraphically, in magnetic and palaeomagnetic characteristics. The average amount of magnetic minerals in tuffs is lower in the upper part of the section (Table 4); $Q_n = 0.09$ is anomalously low (it is likely that at low concentrations of magnetic minerals a contribution of paramagnetic dark-coloured minerals to magnetic susceptibility becomes noticeable). Swarms of unit vectors can be conventionally grouped for directions A and B (Fig. 7, Table 5). For many samples from both groups, the ChRM is essentially stable and presents a single component (Fig. 6i). That is why the great-circle analysis is not effective for these rocks.

7.1.4 Section 11

This section is characterized by high k and NRM values (Table 4) and relatively low ChRM/NRM, which reflects the non-stability of the NRM and results in highly scattered ChRM directions (Fig. 7, Table 5).

Downloaded from https://academic.oup.com/gji/article/130/3/606/674309 by Curtin University Library user on 29 April 2026

Table 4. Some magnetic characteristics of the basalts and tuffs.

Site	Rocks	N	k	NRM	Q_n	ChRM/ NRM	T_d
4	Bas. pebb.	24	56.0	2.17	1.0	0.65	558
			25.6–90.2	0.3–9.6	0.15–4.1	0.1–2.4	520–630
1	Tuffs	32	10.5	0.07	0.3	0.45	580
			0.15–35.0	0.003–0.58	0.04–3.3	0.12–4.1	520–650
2 Upper	Tuffs	19	0.74	0.003	0.09	0.71	540
			0.39–1.9	0.0005–0.01	0.03–0.17	0.2–1.35	520–580
Lower	Tuffs	7	12.9	0.13	0.32	0.74	562
			6.7–18.4	0.026–0.31	0.05–0.6	0.5–1.18	520–600
3	Tuffs	17	16.8	0.2	0.32	0.54	564
			3.1–36.9	0.03–0.46	0.05–0.62	0.08–2.2	520–600,650
5	Tuffs	24	20.7	0.1	0.19	0.46	561
			0.2–66.7	0.05–0.32	0.04–0.54	0.03–0.9	520–600
6	Sill	11	50.5	2.17	1.1	0.64	560
			42.0–63.2	0.6–5.33	0.3–2.6	0.3–1.16	550–580
6	Tuff–1	4	0.22	0.007	0.7	0.64	575
			0.03–0.44	0.001–0.016	0.5–0.9	0.18–1.0	560–600
6	Tuff–2	5	0.4	0.21	1.04	0.3	608
			0.25–0.58	0.13–0.38	0.84–1.3	0.16–0.59	0–620
6	Tuff–3	7	19.5	0.56	0.70	0.63	577
			17.0–21.4	0.38–0.65	0.55–0.87	0.44–1.0	560–610
7	Tuffs	5	15.7	0.18	0.27	0.50	575
			6.3–39.7	0.05–0.34	0.15–0.64	0.07–0.87	560–600
7	Sill	4	27.1	0.82	0.81	0.09	572
			21.2–36.1	0.59–1.06	0.52–1.26	0.07–0.11	560–600
8	Tuffs	3	26.4	0.2(2), 2.47	0.2(2), 2.4	0.58	580
			2.7–31.1			0.38–1.7	560–630
9	Tuffs	8	30.5	0.22	0.2	0.37	570
			7.0–52.5	0.11–0.44	0.1–0.4	0.07–0.57	520–600
10	Tuffs	2	28.3–29.1	0.21–0.24	0.19–0.22	0.66–1.0	560–600
10	Sill	3	41.4	0.47	0.33	0.36	558
			31.2–54.1	0.28–0.81	0.13–0.65	0.26–8.7	520–580
11	Tuffs	14	17.1	0.23	0.33	0.22	560
			4.1–30.7	0.03–0.47	0.18–0.52	0.03–0.7	500–580
12	Tuffs	18	20.3	0.28	0.37	0.46	561
			4.0–42.3	0.026–0.50	0.12–0.79	0.08–1.05	500–600

N is the number of hand samples; k is magnetic susceptibility, 10^{-3} SI units; NRM is natural remanent magnetization intensity, A m^{-1}; ChRM is intensity of the stable part of NRM after heating to 200 °–400 °C; Q = NRM/0.04k is Koenigsberger's ratio; T_d is deblocking temperature, °C. The upper rows refer to average values, lower rows to the limits.

7.2 The middle member

7.2.1 Sections 6–10, tuffs

High magnetic tuffs dominate in sections 7–10 (Table 4). Three horizons can be distinguished in section 6 (Table 4): the uppermost one is in contact with the basalt sill, lying above it. The intrusion resulted in an increased magnetic and palaeomagnetic stability of the rocks. Moving away from the contact, maghemite appears in the middle horizon; its presence is indicated by a 20–50 per cent decrease of *k* after heating of the samples to 300 °–400 °C; the amount of maghemite decreases in the lower horizon (10–20 per cent reduction of *k*, Table 4). Such a regular maghemite distribution with respect to the sill contact can be explained by a heating of 'wet' tufaceous sediments by the intrusion. In the close vicinity of the intrusion, heating was high-temperature (baked contact) and maghemite could not form; in the middle horizon heating was not higher than a few hundred degrees, and these temperature conditions were most favourable for maghemite formation.

The whole zone of heating in section 6 is characterized by A magnetization only (Table 5, Figs 7 and 8e); in the endo- and exocontact of the sill, palaeomagnetic directions coincide (Figs 7d and e). Above and below it, palaeomagnetic directions are more scattered and B magnetization appears (sections 7–9, Figs 7 and 8f). This result can be considered as a positive baked-contact test (Fig. 8).

7.2.2 Sections 6, 7, 10, basalt sills

J_s and *k* of these three sills are typical for basalts: they are high and rather uniform (Tables 3 and 4). At the same time, their NRM and hence Q_n, and ChRM/NRM change regularly: in five samples from sill 6 located near the top (Fig. 8a) and the bottom (Fig. 8d), the ChRM component of direction A of reversed polarity is clearly distinguished, while at the endocontact of sill 7 magnetization A of normal polarity is recognized ($D_a = 73°$, $I_a = 53°$); for other samples from the inner parts of sills, ChRM directions are either anomalous (with mean $D_a = 87°$, $I_a = -61°$, Fig. 8b) or similar to B magnetization (Fig. 8c).

Downloaded from https://academic.oup.com/gji/article/130/3/606/674309 by Curtin University Library user on 29 April 2026

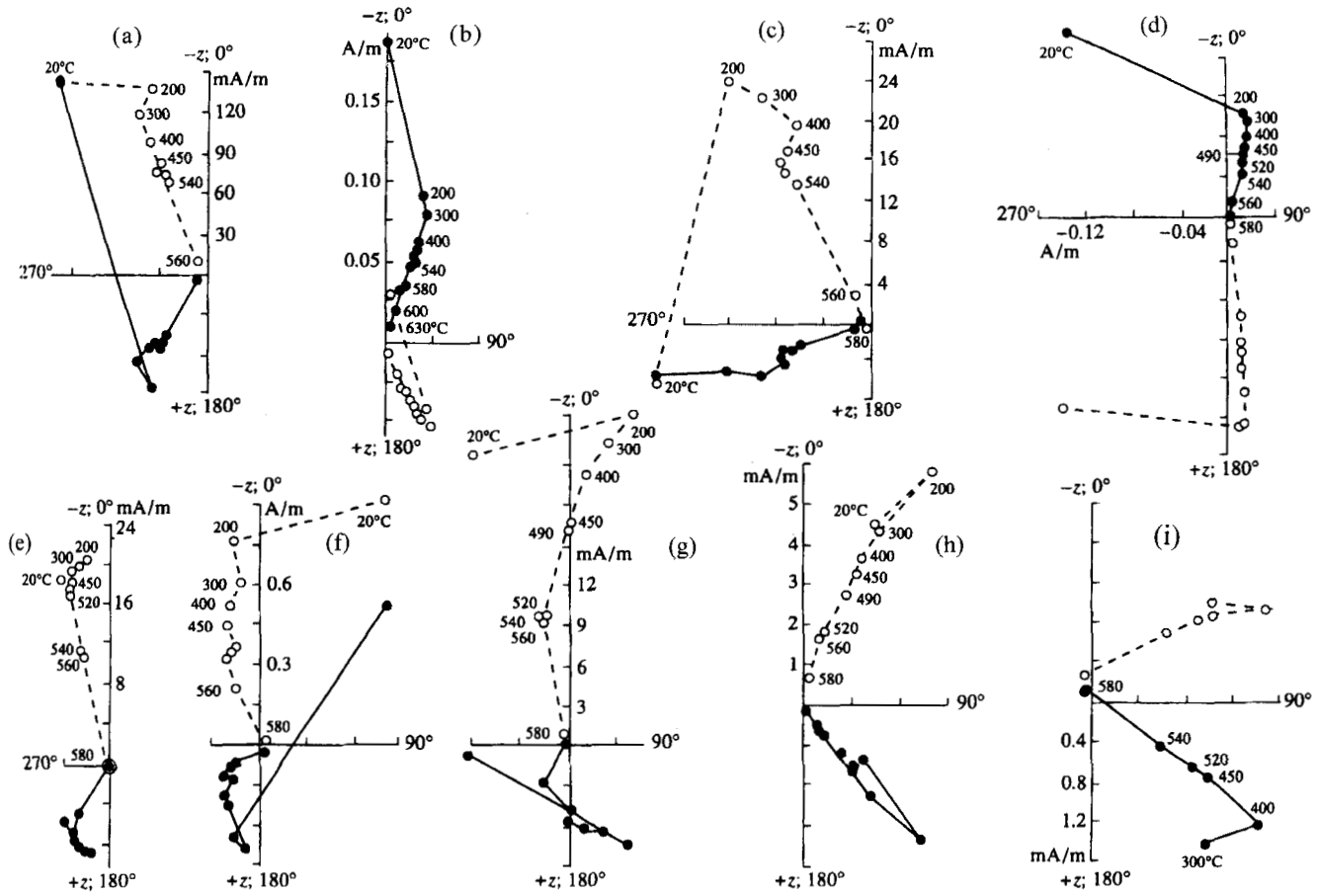


Figure 6. Representative orthogonal plots (Zijderveld's diagrams after tilt correction). (a) Sample Sh125, section 1; (b) L165, section 3 (haematite prevails); (c) L234, section 5; (d) L357, section 12; (e) L117, section 1; (f) L125, section 1; (g) L224, section 5; (h) Sh111, section 1; (i) L152, section 2. (a)–(d) Magnetization A predominates; (e)–(g) both components A and B are present; (h)–(i) magnetization B predominates. Solid circles and solid lines refer to the horizontal projection; open circles and dotted lines, to the vertical one.

Some of the anomalous directions have an NRM stability not worse than samples with only magnetization A, and often $Q_n > 1$, ChRM/NRM = 0.4–1.0. These data and the different magnetic polarities of samples from the endocontacts of sills 6 and 7 lead to the assumption that the geomagnetic field has changed its sign during the ChRM acquisition. The palaeomagnetic stability of basalts is supported by the conglomerate test.

7.2.3 Pebbles of igneous rocks

Magnetic characteristics of pebbles are very similar to those of basalts from the studied sills. Minerals similar to magnetite prevail among magnetic minerals. The presence of haematite is also possible in three pebbles ($T_d > 630\,^\circ$C, Tables 3 and 4). Samples with $Q_n < 0.5$ are characterized by a sharp factor of 2–9 decrease or increase of the NRM intensity after heating to 200 °–300 °C; in contrast, the NRM of the samples with $Q_n = 0.6$–4.1 usually decreases by 5–30 per cent. At the same time, susceptibility does not change. This points to the absence of maghemite as a rule and to the presence of a substantial portion of viscous remanence.

Many directions of the low-temperature component NRM (below 300 °C) are distributed close to the directions of the modern geomagnetic field of normal and reversed polarity and between them (Fig. 9, Table 6). The mean direction of B magnetization of tuffs adjacent to conglomerates lies on the circle that connects the directions of the modern field of normal and reversed polarities. The points located far from this great circle refer to the most stable samples. The orthogonal plots show rectilinear decay of the ChRM above 400 °C to the origin; the distribution of ChRM vectors tends to be random ($K < 1.3$, Table 6); mean ChRM directions are close to the direction of B magnetization (Tables 5 and 6). Pebbles were sampled from intraformational conglomerates, which is why we conclude that igneous rocks in the studied sections should have a ChRM component close to the primary component. Partial remagnetization of rocks by a field with a direction close to B was possible.

7.3 The upper member

This member is characterized only by section 12. It consists of magnetic tuffs (Table 4), with a horizon of low magnetic tuffs of about 5 m thickness in the middle part of the section; $k = (4–10) \times 10^{-3}$ SI units. According to component analysis, a high-temperature ChRM with mainly reversed polarity was distinguished; its direction is close to that of A magnetization (Fig. 7).

Downloaded from https://academic.oup.com/gji/article/130/3/606/674309 by Curtin University Library user on 29 April 2026

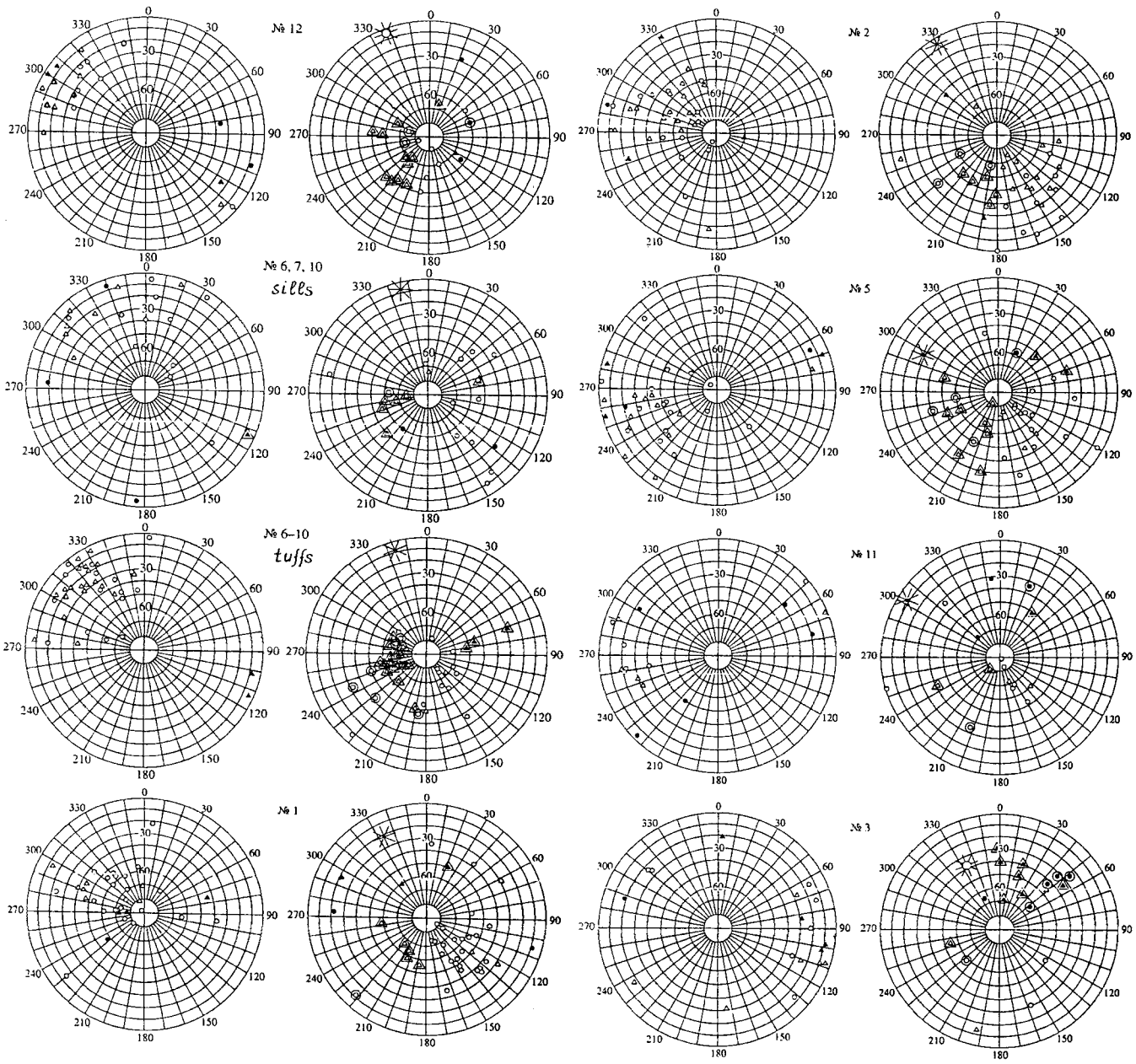


Figure 7. Stereographic projections of ChRM directions of samples for each section. Solid/open symbols denote projection onto the lower/upper hemisphere. Triangles refer to the most stable high-temperature components (ChRM/NRM >0.5; $Q_n>0.3$) and form rectilinear segments going to the origin, circles represent less stable components (ChRM/NRM <0.3) with lower deblocking temperatures. Double symbols refer to directions to the calculated mean-site A directions. The present-day geomagnetic field direction is marked by the large star (in the ancient coordinates). *In situ* data are to the left, and data after tilt correction (the ancient coordinates) are to the right.

7.4 Section 3

Rocks are enriched by magnetic minerals (Table 4); in six samples it is likely that haematite ($T_d>650\,^\circ$C) is present together with magnetite (Tables 3 and 4).

After thermal demagnetization above 300 °C, one component A, mainly prefolding ($K_a/K_c=2$, Table 5) of the normal polarity, was found for most of the samples (Fig. 7). Usually $T_d>560\,^\circ$C, and often it is 580 °C, or even higher than 650 °C; this means that the ChRM component of the A direction is associated with haematite as well (Fig. 6b). Besides A magnetization, in four samples a component B was revealed ($D_a=163\,^\circ$) of reversed polarity and $T_d<560\,^\circ$C, $K_a/K_c=1.57$ (Table 5). Rocks from section 3 have the smallest palaeomagnetic inclination (Table 5), which can be associated with the presence of haematite in these tuffs. However, three of the six haematite-bearing samples have $I=30$–$40\,^\circ$ and three have $I=58$–$69\,^\circ$; five of the eight magnetite-bearing samples have $I=27$–$43\,^\circ$ and three have $I=49$–$66\,^\circ$. This means that the consolidation of sediments did not affect the mean ChRM inclination.

7.5 Long-particle test

For this test, series of orientated tuff samples that have reliably determined A (four samples) or B (five samples) components

Downloaded from https://academic.oup.com/gji/article/130/3/606/674309 by Curtin University Library user on 29 April 2026

Table 5. Site mean directions of ChRM components and great-circle analysis data.

Site	Rocks	A, B	N	D_c	I_c	K_c	D_a	I_a	K_a	α_{95}
12	Tuffs	A	17	302	−15	15.8	230	−69	19.2	7.8
12	Tuffs	Circles	16				36	65		24.7
6+7+10	Sills	Sum	19	345	−22	4.5	86	−83	4.1	15.9
6+7+10	Sills	A	6	311	−19	35.4	254	−65	36.3	9.5
6+7+10	Sills	B?		7	−25	7.0		−61	5.4	17.3
6-10	Tuffs	Sum	34	314	−26	10.6	238	−70	11.2	7.2
6-10	Tuffs	A	29	310	−22	13.4	247	−66	15.8	6.5
6-10	Tuffs	B	14	332	−47	10.2	154	−66	14.3	9.9
7+8	Sum	A	6	310	−19	5.1	256	−63	19.3	13.0
6	Sill	A	5	310	−21	35.0	253	−61	31.0	11.1
6	Tuffs	A	16	318	−20	48.6	269	−69	54.5	4.7
9+10	Sum	A?	8	290	−34	11.5	212	−54	12.3	14.1
6-10	Sum	A		308	−24	40.6	244	−64	98.9	7.0
6-10	Sum	Circles	42			51	69		12.5	
1	Tuffs	Sum	24	202	−61	9.8	160	−63	11.0	8.6
1	Tuffs	A	8	278	−35	12.7	214	−56	23.4	10.2
1	Tuffs	B	24	306	−71	15.4	138	−56	22.6	6.0
1	Tuffs	Circles	23				335	65		10.4
2	Tuffs	Sum	21	297	−18	11.2	154	−57	11.5	9.0
2	Tuffs	A	7	285	−22	9.1	210	−55	22.0	11.3
2	Tuffs	B	21	300	−54	7.7	148	−51	7.9	10.9
2	Tuffs	Circles	21				355	53		2.9
5	Tuffs	Sum	22	251	−27	5.7	197	−65	6.7	11.5
5	Tuffs	A	14	253	−6	10.3	229	−56	10.1	11.8
5	Tuffs	B	14	243	−48	21.8	146	−58	11.2	11.2
5	Tuffs	Circles	21				14	57		15.9
11	Tuffs	Sum	13	258	−18	12.4	183	−63	9.9	12.4
11	Tuffs	A	5	248	−7	7.6	215	−50	13.3	17.2
11	Tuffs	B	7	262	−25	61.7	146	−68	64.5	3.2
11	Tuffs	Circles	14				43	77		>22
3	Tuffs	A	15	90	−6	5.0	29	48	11.2	10.8
3	Tuffs	B	4	295	−16	4.2	163	−43	6.6	27.3
3	Tuffs	Circles	14				62	51		10.0

N is the number of samples or high-temperature NRM components A or B (there are samples with both components A and B, which is why the sum of components A or B may be larger than the sum of samples); 4s = site mean result for four sites 6–10; D and I are palaeomagnetic declination and inclination; K and α_{95} are the parameters of Fisher's statistics; indexes 'c' and 'a' refer to the contemporary (*in situ*) and the ancient (after tilt correction) coordinates 'Circles' are the result of the great circles analysis (Halls 1978).

were selected. Modes of long particles were close to palaeomagnetic declinations for the samples with A magnetization (Figs 10a,b).

For the samples with B magnetization, modes of long-particle directions are similar to the declination of A magnetization and differ greatly from the palaeomagnetic declinations of the samples (Figs 10c,d). Elongated large grains of opaque minerals and silicates usually do not form well-marked modes and their distribution is close to chaotic. Thus, the long-particle test proves the primary detrital nature of A magnetization and the secondary nature of B magnetization.

8 DISCUSSION

(1) As a result of detailed thermodemagnetization and component analysis, high-temperature NRM components of two directions (magnetization A and/or B) were distinguished in all sections. 26 per cent of samples with A magnetization have a normal magnetic polarity and are located in different sections (Fig. 7, Table 5). Parts of orthogonal plots show the rectilinear decay to the origin (Figs 6a–d and 8a,d,e). A positive fold test for some sites (Tables 4 and 6), the conglomerate test for sills (Fig. 9, Table 5), the baked-contact test (Fig. 8) and the long-particle test (Fig. 10) indicate that A magnetization is close to the primary one. Inclinations of A and B magnetizations are close; however, their declinations differ greatly. 96 per cent of samples with B magnetization have a reversed polarity; this means that it was probably formed during one magnetochrone. In some sections, B magnetization is absent or is rather weak (sections 3, 6, 12), while in other sections its contribution is stronger and even dominates (sections 1, 2, 10); in many samples, B magnetization is the only one that is present (except for the viscous magnetization, Figs 6h,i, and 8c). In this case, linear segments on the Zijderveld diagram are not necessarily orientated towards the origin of the coordinates; this means that there are some components that were not distinguished. B magnetization is present in pebbles (Tables 5 and 6); in some sections it was pre-folding and/or synfolding (section 5). In some samples, magnetizations A and B were both found; in these cases, B magnetization usually has lower deblocking temperatures (Figs 6e–g and 8f).

The presence, or moreover the domination, of B magnetiz-

Downloaded from https://academic.oup.com/gji/article/130/3/606/674309 by Curtin University Library user on 29 April 2026

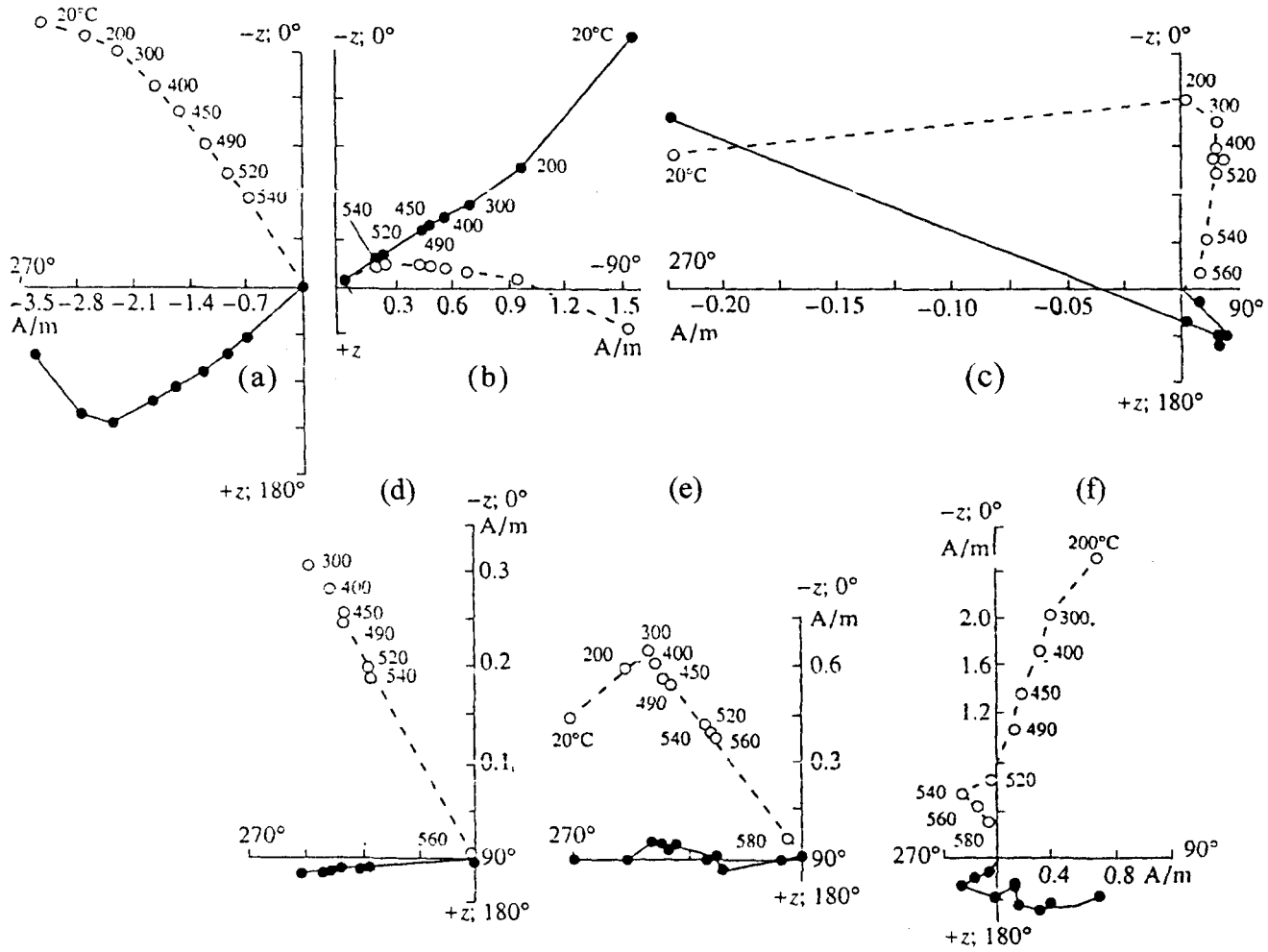


Figure 8. The orthogonal plots demonstrate the positive baked-contact test, Section 6. (a) Sample Sh126: the top of the sill has the only stable ChRM component above 200 °C with direction A. (b) and (c) Samples L285 and Sh159: in the central part of the sill, the viscous remanent magnetization increases, and the direction of the high-temperature component is anomalous (L285) or similar to B (Sh159). (d) L281: near the sill foot, the NRM is the only stable component above 200 °C; its direction is similar to magnetization A. (e) L274: baked tuff near contact with sill; the NRM is the only stable component above 200 °C, and its direction coincides with the direction of the sill sample (d). (f) L323: tuff far from the sill; the proportion of viscous magnetization increases; in the interval 300 °–540 °C the NRM component is close to direction B, and only above 540 °C does a notable component with direction similar to A appear. For designations see Fig. 6.

ation in rocks correlates with relatively low values of Q_n and ChRM/NRM (Fig. 11), i.e. with lower magnetic stability of samples and a larger viscous component. In the sill endo- and exocontact (section 6), B magnetization is absent, but it appears in the inner parts of the sill or far from it (sections 7, 10) where palaeomagnetic stability decreases and the scatter of ChRM directions increases (Fig. 8). The long-particle test for B magnetization is negative (Fig. 10), which allows the B magnetization to be considered as the secondary one. Its origin is unclear; however, B magnetization was formed after 40 °–70 ° clockwise rotation of the block (Table 7).

It is possible that the low effectiveness of the great-circle analysis was caused by the existence of two ChRM components with similar T_d ranges.

Palaeomagnetic directions similar to A and B magnetizations are found that lie between published data for the Upper Cretaceous rocks of the Olutorsk zone and Eastern Kamchatka (Gurevich & Surkis 1994; Savostin & Heifets 1988).

(2) To increase the reliability of the results, the studied sections were gathered into four groups (Table 7). The palaeomagnetic directions of stratigraphically and geographically close sections 1, 2 and 5 were combined. As a result, the pre-folding origin of A magnetization is shown: $K_a/K_c = 14.4$. The mean palaeomagnetic directions for tuffs and sills in sections 6–10 were combined. Because of the monoclinic bedding, the fold test was not effective; however, the reliability of results is high due to the conglomerate and baked-contact tests. The inclination of A magnetization of tuffs from sections 6–10 is 10 ° higher than that of the lower member (Tables 5 and 7). Section 12 cannot be combined with any other. The reliability of the mean direction obtained for this section is supported by its similarity with palaeomagnetic directions A for the sections of the middle member, where reliability was proved by several tests. The absence of B magnetization indicates that the tuffs of this section were not remagnetized. In the lower part of the general sequence, sections 3 and 11 were combined. The ChRM of section 3 has the A directions of normal polarity; samples from section 11 have both the reversed and normal magnetic polarities (Fig. 7). A difference in dominating magnetic polarities in sections 3 and 11 indicates that they could have been

Downloaded from https://academic.oup.com/gji/article/130/3/606/674309 by Curtin University Library user on 29 April 2026

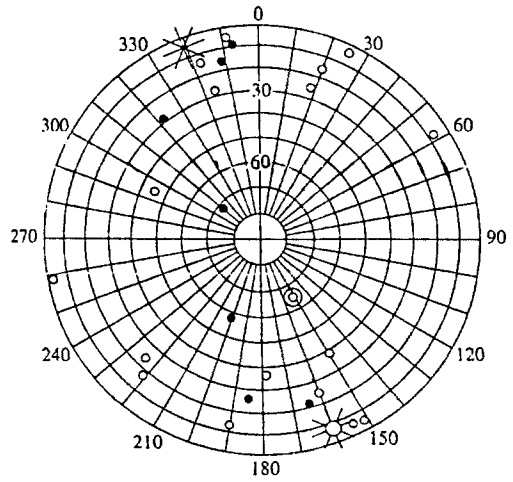


Figure 9. The position of difference vectors (between the initial NRM and after thermotreatment at 200 °–300 °C) for pebbles from intraformational conglomerate after tilt correction (the ancient coordinates). Closed/open circles represent projections onto the lower/upper hemisphere. The present-day geomagnetic field direction of the normal and reversed polarity is marked by stars. The mean direction of magnetization B for tuffs of sections 6–10 is marked a double circle.

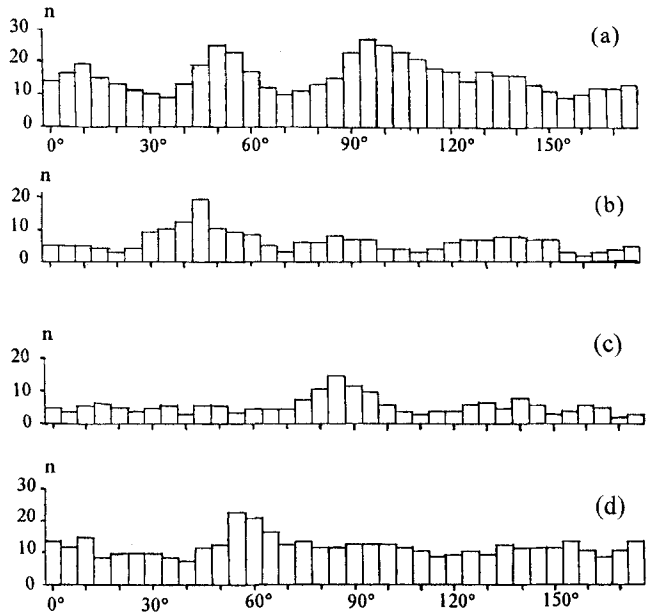


Figure 10. Histograms showing directions of the long axes of opaque mineral grains ('long particles'). (a) Three samples from section 5, magnetization A, mean $D_a = 104°$. Two modes of long particles are 50° and 95°. (b) One sample from section 3, magnetization A, $D_a = 76°$. The mode of long particles is 45°; (c) Two samples from section 1, magnetization B, mean $D_a = 314°$. The mode of long particles is 85°; (d) Three samples from section 11, magnetization B, mean $D_a = 333°$. The mode of long particles is 55°.

formed asynchronously, although they must be close in age because of the similarity of their mean palaeomagnetic directions. Within the confidence angles, palaeomagnetic directions of the normal and reversed polarities in these sections are antiparallel in the ancient coordinate system: $D_a = 36°$ and 231°, $I_a = 46°$ and $-54°$, $\alpha_{95} = 9.2°$ and 10.8°; this means that the reversal test is positive. The magnetic properties of samples with normal and reversed magnetizations do not differ. Summation of palaeomagnetic directions for sections 3 and 11 proves that the summed direction is prefolding ($K_a = 674$ and $K_c = 14.7$); a similar result was obtained by the Holls method (Table 7). The mean palaeomagnetic direction of A magnetization for sections 3 and 11 can be considered as close to the primary one.

(3) If sections are placed so that the inclinations of A magnetization increase (Tables 5 and 7), their sequence will mainly coincide with the stratigraphic interpretation: moving from sections 1, 2, 5 (the lower member) to section 12 (the upper member), the mean inclination increases from $I_a = 56°$ to 69°; it corresponds to an increasing palaeolatitude from 37 °N to 53 °N (Table 7). The range of palaeolatitude where the Khapitsky Formation was formed may be increased if we take into account the data for sections 11 and 3, where the mean inclination of A magnetization is 49° and, hence, the palaeolatitude is 30 °N (Table 7).

Thus, the results indicate that the Upper Cretaceous–Lower Palaeocene volcanic zone has undergone absolute displacement during its formation; however, estimates of drifting velocities require special discussion.

(4) An estimate of the meridional component of the drifting rate of the volcanic arc, where the Khapitsky Formation was accumulated, depends on evaluations of both the palaeolatitudes of the studied sections and their ages. Our relatively conventional age estimates are based on magnetostratigraphical allocation of the sections to the Upper Campanian, Maastrichtian and Lower Palaeocene times (the time interval when the entire Khapitsky Formation was accumulated). In this interval, especially in its earlier part, normal polarity prevails. An allocation of all sections characterized by reverse polarity to one of the relatively short chrons would have

Table 6. Results of the pebbles demagnetization.

NRM component, temperature of heating		N	D_a	I_a	K	α_{95}
1 component,	20–300 °C	24	8	−6	1.0	27.7
2 component,	>400 °C	24	126	−42	1.3	25.1
	400 °C	24	132	−33	1.3	25.2
	520 °C	24	128	−29	1.2	25.7
	560 °C	24	128	−10	1.1	27.9
	600-630 °C	6	131	56	1.3	49.6
Circles, component 1		15	350	−6	2.6	22.4
Circles, component 2		24	83	11		>90

For explanation, see Table 5.

Downloaded from https://academic.oup.com/gji/article/130/3/606/674309 by Curtin University Library user on 29 April 2026

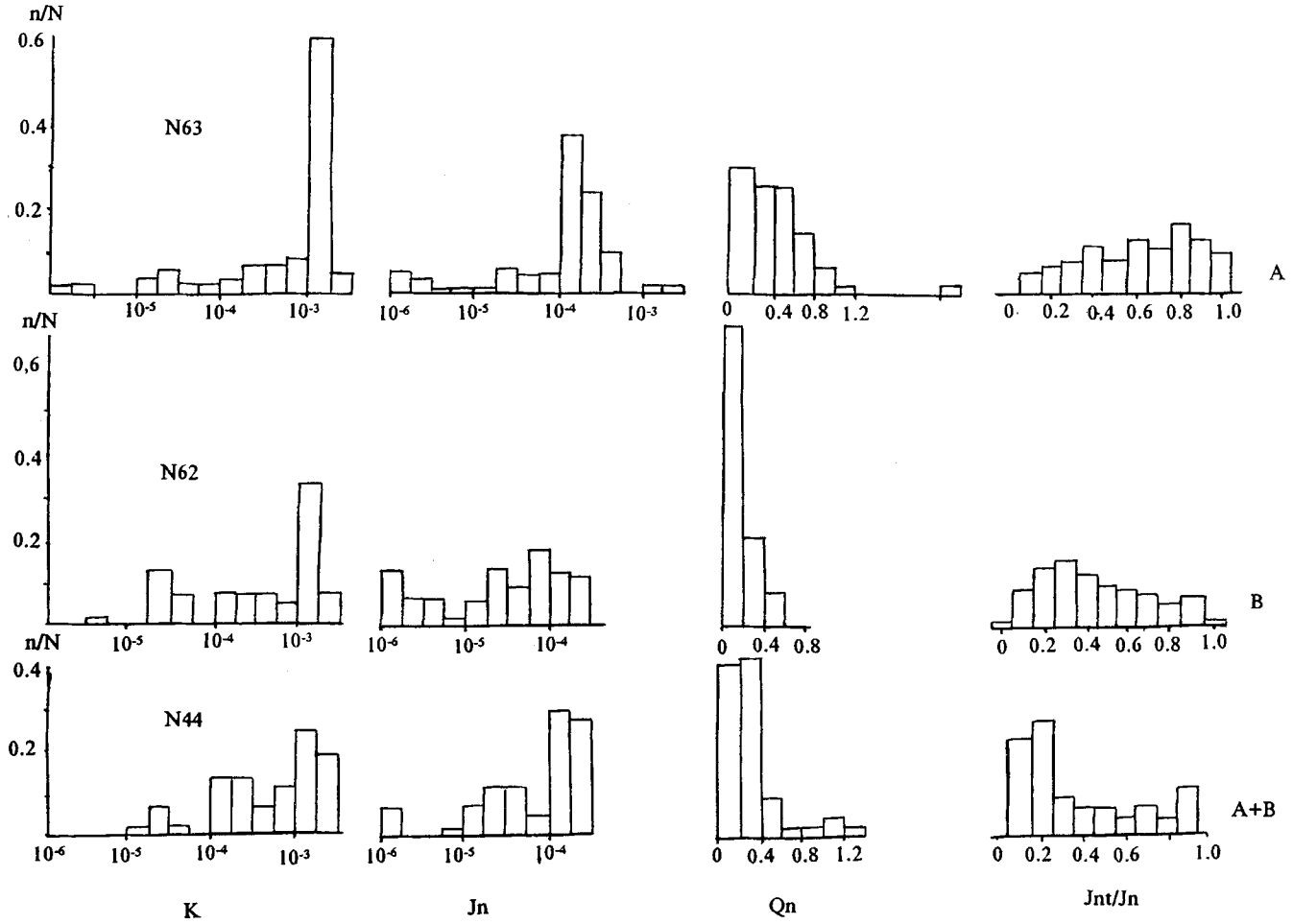


Figure 11. Distributions of magnetic susceptibility k, natural remanent magnetization NRM, Königsberger's ratio Q_n and the ratio of the NRM intensity after 200°–300 °C thermocleaning (ChRM) to the initial NRM intensity for tuff samples with magnetization A (top row of histograms), B (middle row), A + B (bottom row).

Table 7. Generalized palaeomagnetic directions of the magnetization A.

Sections	n	D_c	I_c	K_c	D_a	I_a	K_a	α_{95}	P	PL
12	1/17	302	−15	15.8	230	−69		7.8	R	53
12, circles	/16				36	65		24.7		47
6-10	5/	311	−21	1330	251	−66	1395	2.7	R	48
6-10,circles	/42				51	69		12.5		52
1+2+5	3/	271	−22	14.2	218	−56	205	5.6	R	37
1+2+5,circl.	/60				357	62		11.0		43
3+11	2/	81	−1	14.7	32	49	674	3.8	N,R	30
3+11,circles	/26				54	52		12.3		33

N is the number of sites/samples; P denotes predominant magnetic polarity (N = normal, R = reversal); PL is a palaeolatitude. For other designations, see Table 5.

resulted in an improbably high value of the drifting rate. This is why the sections were assumed to be associated with different chrons: section 12 with the time interval that includes chron 27r and the lower Palaeocene part of chron 26r; sections 6–10 with chron 29r; sections 1,2 and 5 with chron 31r. Section 11, which includes samples of both polarities, was assumed to belong to chron 32r, where a short interval of normal polarity has existed. Section 3 was assumed to belong to the Campanian chron 33n. Correspondingly, an estimation of the palaeolatitude based on a joint interpretation of sections 3 and 11 refers to the time interval which includes chrons 32r and 33n (Fig. 12). As intervals that match the uppermost and lower sections are relatively long (3.5 and 6 Myr, respectively), the difference between the maximal (19 Myr) and minimal (9.5 Myr) estimates for the duration of the studied part of the Khapitsky Formation accumulation is significant.

The difference between the maximal (38.5 °) and minimal (11.5 °) estimates of the value of the northward drift of the arc

Downloaded from https://academic.oup.com/gji/article/130/3/606/674309 by Curtin University Library user on 29 April 2026

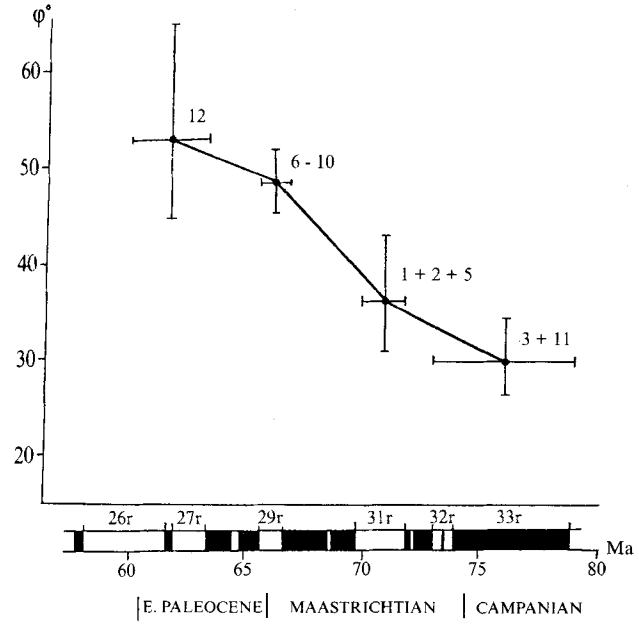


Figure 12. Palaeolatitude versus age plot of our results. Geomagnetic polarity timescale after Cande & Kent (1992).

is even greater; it is caused by the large errors in the determination of the palaeolatitudes for sections 3 and 11, and especially for section 12 (Fig. 12). This is why palaeomagnetic estimates of the meridional component of the average drifting rate of the Achaivayam–Valaginsky arc lie in a wide range, from 6 to 45 cm yr^{-1}. Drifting velocities that correspond to the upper part of this range are unlikely; however, even the smallest of the possible estimates supports the hypothesis of a fast drift of the active arc together with the subduction zone. At the same time, our estimates prove that the arc, where the Khapitsky Formation was accumulated, was formed on one of the plates of the Pacific Ocean drifting rapidly northwestwards. If we take the average values of the age and palaeolatitude estimates for the considered sections as the basic, then during 15 Ma the Achaivayam–Valaginsky arc has moved northwards by 23°, with an average drift rate of about 17 cm yr^{-1}. This estimate coincides to within an order of magnitude with the meridional component of the absolute rate of the northwestern part of the Kula plate in Maastrichtian and Lower Palaeocene times, which is estimated to be 12 cm yr^{-1} according to the data of Engebretson, Cox & Gordon (1985). The corresponding rate for the Pacific plate was half that at that time.

(5) Data on the Eurasian apparent polar path (Khramov 1991) and absolute displacements of the marginal plates of the northern Pacific (Engebretson, Cox & Gordon 1985) show that from the second half of the Campanian until the middle Palaeocene, the margin of Eurasia where the studied sites of the Kumroch Ridge are now located has drifted from 70°N to 62°N. Hence, at the beginning of the Khapitsky Formation deposition, the island arc was located approximately 40° to the south of the Eurasian continent and only 8°–10° to the south by the time of the upper-member formation. If the arc continued to drift towards the continent with the same rate (17 cm yr^{-1} along the meridian), then over 4–5 Myr it would have approached the continent, so it would have been possible for terrigeneous flysch material from the continental source (the Drozdovsky and Talnikovsky formations) to accumulate there. At the same time, volcanic activity stopped; this could have been caused by the initial collision of the former arc and the continent, which would have led to the reconstruction of plate boundaries and the formation of the Aleutian arc.

(6) The mean palaeomagnetic declination of the Khapitsky Formation is $46° \pm 14°$, which differs greatly from the declination calculated for the region of study using the position of the Eurasian pole ($D = 354°$, Khramov 1991). It is likely that this discrepancy is associated with the clockwise rotation of the plate during the collision of the former island arc with the continent. A large value of the angular standard deviation (14°) is probably due to local rotations of small blocks within the plate.

9 CONCLUSIONS

The palaeomagnetic study of the Upper Cretaceous–Lower Palaeocene volcanic-sedimentary sequence of the Kumroch Ridge reveals that, during the formation of Achaivayam–Valaginsky arc, the arc and the subduction zone related to it have undergone absolute northward displacement. According to the average palaeoinclinations for the studied horizons of the Khapitsky Formation, the displacement was about 2500 km. Stratigraphic and palaeomagnetic estimates of the average duration of the volcanic accumulation (about 15 Myr) give 17 cm yr^{-1} for the meridional component of the rate of the absolute drift of the Late Cretaceous–Early Palaeocene subduction zone. This value is in good agreement with the submeridional rate of the Kula plate displacement at the same time.

ACKNOWLEDGMENTS

We thank N. M. Levashova and G. N. Novikov (Geological Institute, Russian Academy of Sciences) who participated in sampling, Yu. S. Bretshtein and G. S. Yanova, who made some laboratory measurements, V. A. Tselmovich for the electron microprobe analysis, and M. Y. Boyarinova and other participants of the Kronotsk field group of the Central Complex Expedition of 'Kamchatgeologia' for their support in the field work. We wish to thank E. Gurria for revising the language. Dr Jason Ali read an early version of the manuscript, and his comments were most helpful. The study was supported by the Russian Funding for Basic Research (grant 93–05–8615).

REFERENCES

Alekseev, E.S., 1979. The main features and structures of the southern part of the Koryak highland, *Geotektonika,* **11,** 85–95 (in Russian).

Bakhteev, M.E., Beniamovsky, V.N., Bragin, N.Yu., Vitukhin, D.I., Morozov, O.A., Sinelnikova, V.N., Tikhomirova, S.R. & Shantser, A.E, 1994. New data about stratigraphy of Mesozoic-Cenozoic of Eastern Kamchatka (Valaginsky Ridge), *Stratgrafiya i geologicheskaya korrelyatsia,* **2(4),** 77–84 (in Russian).

Cande, S.C. & Kent, D.V., 1992. A new geomagnetic polarity time scale for the Late Cretaceous and Cenozoic, *J. geophys. Res.,* **97,** 13917–13951.

Cox, A. & Hart, B., 1986. *Plate Tectonics: How it Works,* Blackwell Scientific Publications, Oxford.

Engebretson, D.C., Cox, A. & Gordon, R.G., 1985. Relative motions between oceanic and continental plates in the Pacific basin, *Geol. Soc. Am. Spec. Paper,* **206.**

Flyorov, G.B. & Koloskov, A.V., 1976. *Alkali-basalt Magmatism of Central Kamchatka,* Nauka, Moscow (in Russian).

Downloaded from https://academic.oup.com/gji/article/130/3/606/674309 by Curtin University Library user on 29 April 2026

Garfunkel, Z., Anderson, C.A. & Shubert, G., 1986. Mantle circulation and the lateral migration of subducted slabs, *J. geophys. Res.*, **91,** 7205–7224.

Gurevich, E.L. & Surkis, Ju.F., 1994. Palaeomagnetic estimation of the horizontal tectonic movements of the Olutorsko-Kamchatka zone, *Fizika Zemli,* **10,** 58–69 (in Russian).

Halls, H.C., 1978. The use of converging remagnetization circles in palaeomagnetism, *Phys. Earth planet. Inter,* **16,** 1–11.

Kamenetsky, V.S., Sobolev, A.V., Karpenko, S.F. & Portniagin, M.V., 1993. Ultramafic volcanism of Eastern Kamchatka: association, geochemistry and petrology, in *Accretionary Tectonics of Eastern Kamchatka,* pp. 156–196, Nauka, Moscow (in Russian).

Khramov, A.N., 1991. Standard rows of palaeomagnetic poles for the plates of Northern Eurasia: correlation with the problems of palaeogeodynamics of the USSR territory, in *Palaeomagnetism and Palaeogeodynamics of the USSR Territory,* pp. 135–149, VNIGRI, Leningrad (in Russian).

Kononov, M.V., 1989. *Plate tectonics of the North-Western Pacific,* Nauka, Moscow (in Russian).

Konstantinovskaia, E.A., Zinkevich, V.P., Tsukanov, N.V. & Garanina, S.A., 1993. Composition, structure and formation of the Upper Cretaceous—Lower Palaeogenic complexes of East Ridges of Kamchatka, in *Accretionary Tectonics of Eastern Kamchatka,* pp. 59–113, Nauka, Moscow (in Russian).

Kovalenko, D.V., 1990. Palaeomagnetic studies of the island arc complexes of the Karaginsky island and the Malinovsky ridge and tectonic interpretation of results, *Geotectonika,* **2,** 92–101 (in Russian).

Kovalenko, D.V., 1992. Palaeomagnetic and geologic data analysis for the Olutorskaia zone and the Karaginsky island region: tectonic implications, *Geotectonika,* **5,** 78–96 (in Russian).

Magakian, R., Kolesov, G.M., Romashova, T.V. & Konstantinovskaia, E.A., 1993. Geochemical peculiarities of the Cretaceous Eastern Kamchatian island arc volcanism, in *Accretionary Tectonics of Eastern Kamchatka,* pp. 114–155, Nauka, Moscow (in Russian).

Nishitani, T. & Kono, M., 1983. Curie temperature and lattice constant of titanomagnetite, *Geophys. J. R. astr. Soc.,* **74,** 585–600.

Pechersky, D.M., 1970. The main ideas and experimental testing of 'long particles' method, *Izvestia AN SSSR, Ser. Geol.,* **3,** 103–111 (in Russian).

Pechersky, D.M., Didenko, A.N., Lykov, A.V. & Tikhonov, L.V., 1993. Petromagnetism of the oceanic lithosphere, *Fizika Zemli,* **12,** 17–45.

Review of the tectonic evolution of Kamchatka, 1987. Nauka, Moscow (in Russian).

Savostin, L.A. & Heifets, A.M., 1988. Palaeomagnetism of the Maastrichtian-Palaeocenian island arc formations in the Olutorskaia zone, in *Palaeomagnetism and Accretion Tectonics,* pp. 127–140, Trudy VNIGRI, Leningrad (in Russian).

Shantser, A.E., Shapiro, M.N., Koloskov, A.V., Chelebaeva, A.I. & Sinelnikova, V.N., 1985. Structural evolution of the Lesnovsky uplift and adjacent area in Cenozoic (Northern Kamchatka), *Tikhookeanskaya Geologia,* **4,** 66–74 (in Russian).

Shapiro, M.N., 1995. Late Cretaceous Achaivaiam-Valaginskaia volcanic arc and some problems of the Northern Pacific plate kinematics, *Geotectonika,* **1,** 58–70 (in Russian).

Tsukanov, N.V., 1991. *Tectonic Evolution of the Kamchatka Seaside Zone in the Late Mesozoic–Early Cenozoic,* Nauka, Moscow (in Russian).

Zinkevich, V.P. & Tsukanov, N.V., 1993. Formation of the accretionary structure of Eastern Kamchatka, in *Accretionary Tectonics of Eastern Kamchatka,* pp. 249–258, Nauka, Moscow (in Russian).

Downloaded from https://academic.oup.com/gji/article/130/3/606/674309 by Curtin University Library user on 29 April 2026